

Virtualized Lustre

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Versions

steps in this guide should work for these versions of software

Virtualbox	6.0.20
Centos	7.8
Lustre	2.12.5
SPL	0.7.13
ZFS	0.7.13

Check the lustre support matrix to be sure you have the right versions whenever you build a lustre environment.

<https://wiki.whamcloud.com/display/PUB/Lustre+Support+Matrix>

VM BUILD

Download Centos 7 minimal ISO from centos.org

Create a new VM

Administrator: admin

Give it 2 GB ram and an 8GB vdi hd

Mount the centos 7 image to the optical drive

Set up 2 nics

1. NAT
2. Host Only (should be vboxnet0). Assign 192.168.56.0/24

Disable firewall and selinux

```
systemctl disable firewalld
#verify with:
systemctl status firewalld
```

Disable selinux:

In /etc/selinux/config

SELINUX=disabled

```
#verify with:
sestatus
```

Configure network to start properly every time you boot.

Disable NetworkManager

```
systemctl disable NetworkManager
```

Manually configure your network adapters

Example:

enp0s8 /etc/sysconfig/network-scripts/ethcfg-enp0s8

```
BOOTPROTO=none
ONBOOT=yes
PREFIX=24
IPADDR=192.168.56.x
```

PREPARE FOR LUSTRE COMPILE

Set up a directory tree:

```
mkdir -p /root/lustre/rpms/{lustre-client,lustre-server,spl-zfs}
mkdir -p /root/lustre/source/{lustre,spl,zfs}
```

Update and Install dependencies for compiling Lustre/ZFS:

```
yum update
yum install asciidoc audit-libs-devel automake bc binutils-devel \
bison device-mapper-devel elfutils-devel elfutils-libelf-devel expect \
flex gcc gcc-c++ git glib2 glib2-devel hmacalc keyutils-libs-devel \
krb5-devel ksh libattr-devel libblkid-devel libselinux-devel libtool \
libuuid-devel libyaml-devel lsscsi make ncurses-devel net-snmp-devel \
net-tools newt-devel numactl-devel parted patchutils pciutils-devel \
perl-ExtUtils-Embed pesign python-devel redhat-rpm-config rpm-build \
systemd-devel tcl tcl-devel tk tk-devel wget xmlto yum-utils zlib-devel\
libaio-devel kernel-devel-$(uname -r) kernel-headers-$(uname -r)
```

If the update install a new kernel, you will need to reboot and install the appropriate kernel-devel and kernel-headers for the running kernel.

Compiling

spl

```
git clone https://github.com/zfsonlinux/spl.git /root/lustre/source/spl
cd /root/lustre/source/spl
```

```
git checkout spl-0.7.13
```

```
sh autogen.sh
./configure --with-spec=redhat
make pkg-utils pkg-kmod
yum localinstall *$(uname -p).rpm
mv *$(uname -p).rpm /root/lustre/rpms/spl-zfs
```

zfs

```
git clone https://github.com/zfsonlinux/zfs.git /root/lustre/source/zfs
cd /root/lustre/source/zfs
```

```
git checkout zfs-0.7.13
```

```
sh autogen.sh
./configure --with-spec=redhat
make pkg-utils pkg-kmod
yum localinstall *$(uname -p).rpm
mv *$(uname -p).rpm /root/lustre/rpms/spl-zfs
```

Lustre

```
git clone git://git.whamcloud.com/fs/lustre-release.git /root/lustre/source/lustre
cd /root/lustre/source/lustre
git checkout 2.12.5
sh autogen.sh
#SERVER
./configure --enable-server --disable-ldiskfs --with-zfs --with-spl --with-linux=/usr/src/kernels/$(uname -r)
make rpms
mv *.rpm /root/lustre/rpms/lustre-server
make distclean

#CLIENT
./configure --disable-server --enable-client --with-linux=/usr/src/kernels/$(uname -r)
make rpms
mv *.rpm /root/lustre/rpms/lustre-client
```

Create Lustre Cluster VMs

Clone gold image using full clone rather than linked clone.

Create the following clones:

Set the IP for each client as listed. If your IP range is not 192.168.56, change the numbers below to match your IP range.

Add 2x5GB hds to mds00, 1x5GB hd to mds01, and a 10GB hd to oss00 and oss01

```
mds00 192.168.56.10
mds01 192.168.56.20
oss00 192.168.56.30
oss01 192.168.56.40
client 192.168.56.50
```

Mds00 is your mgs and mds server.

Mds01 is your secondary mds

Oss00 and Oss01 are the oss servers on which you will set up your ost volumes.

The client is what you will use to mount lustre and test.

Format and Mount Lustre

Install RPMs

On mds00, mds01, oss00 and oss01:

```
cd /root/lustre/rpms/spl-zfs
yum localinstall -y *.rpm
cd /root/lustre/rpms/lustre-server
Yum localinstall -y *.rpm
```

On client

```
cd /root/lustre/rpms/lustre-client
yum localinstall -y *.rpm
```

LNET configuration

Edit /etc/modprobe.d/lustre.conf on each node:

```
Options lnet ip2nets="tcp0(enp0s8) 192.168.56.[10,20,30,40,50]"
```

Load the Lustre modules

```
modprobe zfs
modprobe lustre

# verify success:
lsmod | grep zfs
lsmod | grep lustre
```

Format

mds00

```
zpool create lustre-mgs /dev/sdb
zpool create lustre-mds00 /dev/sdc
genhostid
mkfs.lustre --mgs --backfstype=zfs --fsname=lustre lustre-mgs/mgt
mkfs.lustre --mdt --mgsnode=192.168.56.10@tcp:192.168.56.20@tcp --failnode=192.168.56.20@tcp --backfstype=zfs --fsname=lustre --index=0 lustre-mds00/mdt0

mkdir -p /mnt/lustre/{mgt,mds00.mdt0,mds01.mdt0}
mount -t lustre lustre-mgs/mgt /mnt/lustre/mgt
mount -t lustre lustre-mds00/mdt0 /mnt/lustre/mds00.mdt0
```

mds01

```
zpool create lustre-mds01 /dev/sdb
genhostid
mkfs.lustre --reformat --mdt --mgsnode=192.168.56.10@tcp:192.168.56.20@tcp --failnode=192.168.56.10@tcp --
backfstype=zfs --fsname=lustre --index=1 lustre-mds01/mdt0

mkdir -p /mnt/lustre/{mgt,mds00.mdt0,mds01.mdt0}
mount -t lustre lustre-mds01/mdt0 /mnt/lustre/mds01.mdt0
```

oss00

```
zpool create lustre-oss00 /dev/sdb
genhostid
mkfs.lustre --reformat --ost --backfstype=zfs --fsname=lustre --index=0 --mgsnode=192.168.56.10@tcp:
192.168.56.20@tcp --failnode=192.168.56.40 lustre-oss00/ost0

mkdir -p /mnt/lustre/{oss00.ost0,oss01.ost0}
mount -t lustre lustre-oss00/ost0 /mnt/lustre/oss00.ost0
```

oss01

```
zpool create lustre-oss01 /dev/sdb
genhostid
mkfs.lustre --reformat --ost --backfstype=zfs --fsname=lustre --index=1 --mgsnode=192.168.56.10@tcp,
192.168.56.20@tcp --failnode=192.168.56.30 lustre-oss01/ost0

mkdir -p /mnt/lustre/{oss00.ost0,oss01.ost0}
mount -t lustre lustre-oss01/ost0 /mnt/lustre/oss01.ost0
```

mount -t lustre is the command that starts the lustre file services. Simply mounting the storage as ldiskfs or zfs will mount the target, but will not trigger the Luster kernel process startup.

To verify that your cluster is running, type the following commands:

```
[root@mds00 ~]# df -hT
lustre

Filesystem      Size  Used Avail Use% Mounted
on
lustre-mgs/mgs  4.8G  2.8M  4.8G   1% /mnt/lustre
/mgt
lustre-mdt0/mdt0 4.8G  3.0M  4.8G   1% /mnt/lustre/mdt

[root@mds00 ~]# lctl
dl

  0 UP osd-zfs MGS-osd MGS-osd_UUID
4
  1 UP mgs MGS MGS
4

  2 UP mgc MGC192.168.56.10@tcp 38603054-2ccc-4
4
  3 UP osd-zfs lustre-MDT0000-osd lustre-MDT0000-osd_UUID
6
  4 UP mds MDS MDS_uuid
2

  5 UP lod lustre-MDT0000-mdtlov lustre-MDT0000-mdtlov_UUID
3
  6 UP mdt lustre-MDT0000 lustre-MDT0000_UUID
4
  7 UP mdd lustre-MDD0000 lustre-MDD0000_UUID
3
  8 UP qmt lustre-QMT0000 lustre-QMT0000_UUID
3
  9 UP lwp lustre-MDT0000-lwp-MDT0000 lustre-MDT0000-lwp-MDT0000_UUID
4
```

Lustre Service Configuration

On mds00,mds01,oss00,oss01 edit/create /etc/ldev.conf

```
mds00 mds01 MGS          zfs:mds00/mgs
mds00 mds01 lustre-MDT0000 zfs:mds00/mdt0
mds01 mds00 lustre-MDT0001 zfs:mds01/mdt0
oss00 oss01 lustre-OST0000 zfs:oss00/ost0
oss01 oss00 lustre-OST0001 zfs:oss01/ost0
```

Lustre can be started by running the following on each server node:

```
service lustre start
```

and stopped by running the following on each server node:

```
service lustre stop

# remove modules if neccessary
lustre_rmmmod
```

Testing

Mount Lustre on the client

```
mkdir /mnt/lustre
mount -t lustre 192.168.56.10@tcp:192.168.56.20@tcp:/lustre /mnt/lustre
```

Write some data

```
dd if=/dev/zero of=/mnt/lustre/foo bs=1024k count=100
```

Lustre Startup & Shutdown (no service)

Startup sequence

import MGT on MDS00 and mount the dataset

```
zpool import lustre-mgs
mount -t lustre lustre-mgs/mgt /mnt/lustre/mgt
```

import and mount MDTs on both MDSs, example:

```
zpool import lustre-mds00
mount -t lustre lustre-mds00/mdt0 /mnt/lustre/mds00.mdt0
```

import OST zpools on both OSSs and mount dataset, example:

```
zpool import lustre-oss00
mount -t lustre lustre-oss00/ost0 /mnt/lustre/oss00.ost
```

mount client

```
mount -t lustre 192.168.56.10@tcp:/lustre /mnt/lustremnt
```

Shutdown sequence

unmount all clients

```
umount /mnt/lustremnt
```

unmount MDT datasets from all MDSs and export MDT zpools, example:

```
umount /mnt/lustre/mds00.mdt0
zpool export lustre-mds00
```

unmount OST datasets from all OSSs and export OST zpools, example:

```
umount /mnt/lustre/oss00.ost0
zpool export lustre-oss00
```

unmount MGT dataset from MDS and export MGT zpool

```
umount /mnt/lustre/mgtzpool
export lustre-mgs
```

Shutdown VMs